COSMOS Cluster 5 Final Project

Lego Submarine

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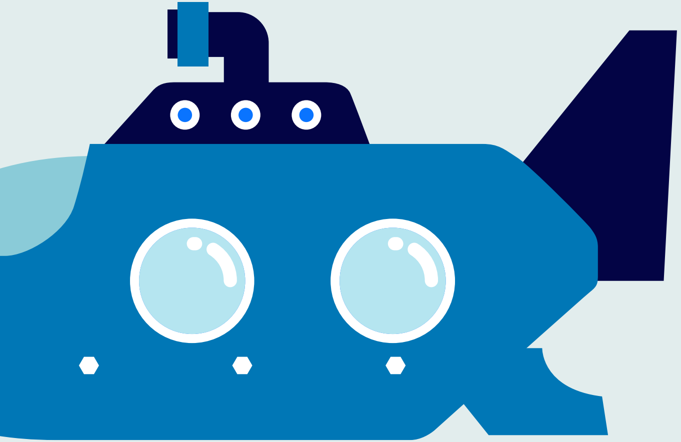
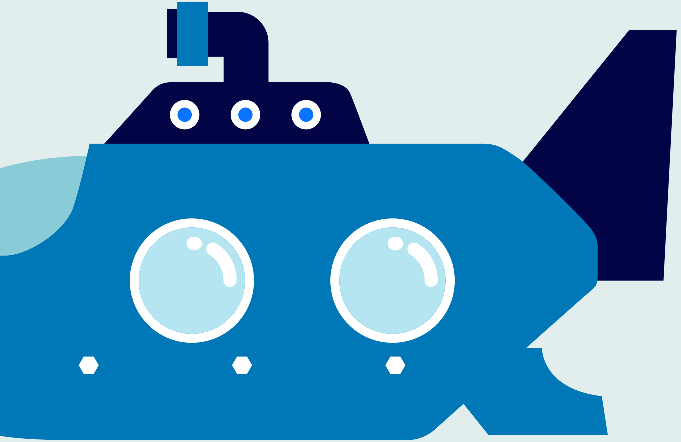


Ideas for Final Project

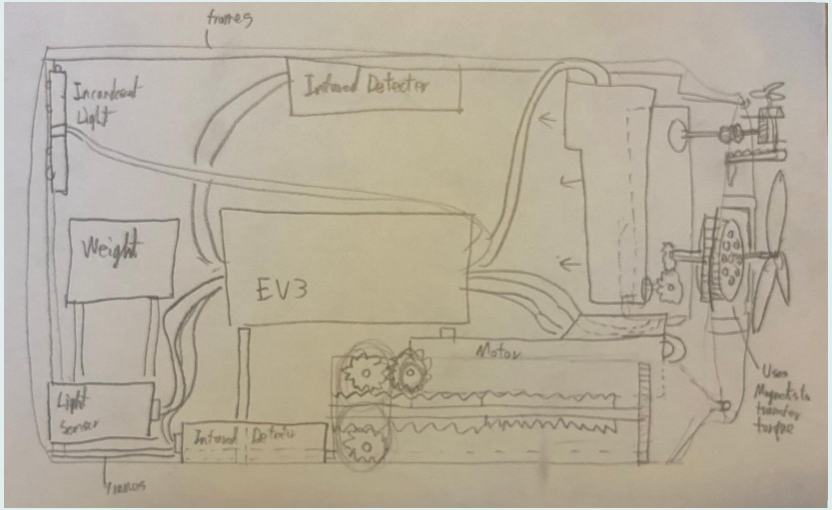
- Autonomous Car
- Lego Drone
- Lego Ship
- Lego Submarine

Why Submarine?



- Challenge
- Saw an example of functional lego submarine

Brainstorm - Structure



- Pump
  - Balloon or Syringe
- Propellers
  - Magnetic

Brainstorm - Functions

- Infrared Sensor
  - Depth Control
- Light Sensor
  - Make it autonomous
  - Flash light

Main Problems:

Jar opening is too small

Either have to construct the submarine in the jar or complete the build and put it in

Water-Proof

Propellers

Solution: Magnets

Lid

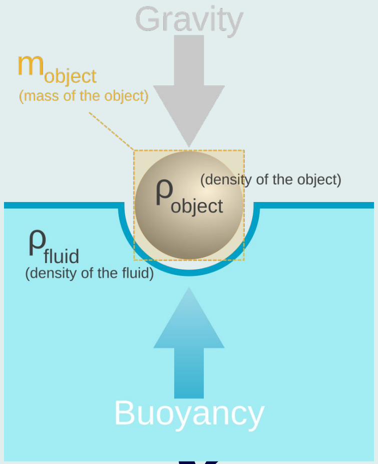
Solution: O'Rings & glue gun

Propellers

Magnets:

- Neodymium magnets in gears
- Weak magnets -> slowly increase speed

Propellers:



Density

Perfect density so that the syringe can control the depth of the submarine

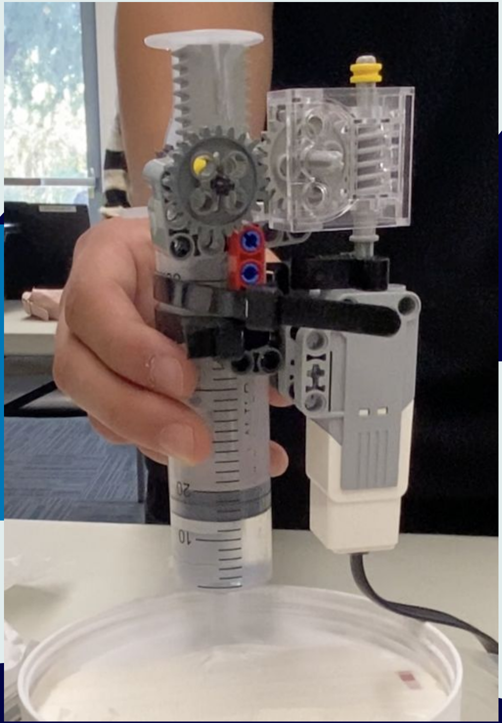
$$\textit{Specific Gravity} = \frac{\text{density of the object}}{\text{density of water}} = \frac{P\ \text{object}}{P\ H_2O}$$

Weights

01
Steel balls
Inside the submarine

02
Pebbles
Outside the submarine

03
Hammer heads
Outside the submarine



Syringe Pump System

- Motor controls the syringe movement
- Pulls in water to increase density
- Pushes out water to decrease density

Syringe Pump Code

```
int dist = 50;
int current = SensorValue[irSensor]

while (current > dist)
{
  setMotorSpeed(Syringe, -100);
}
setMotorSpeed(Syringe,0);

setMotorSpeed(Syringe,100);
sleep(10000);
```

Syringe will pull in water until the depth is above 50 cm

Syringe movement will now take out water to go up

Object Detection

```
task main()
{
  int val = SensorValue[objdet];
  bool stap = false;
  while (true)
  {
    if (val<10) //go straight
    {
      for(int i=15; i<=30; i++)
      {
        if (SensorValue[objdet] > 10)
        {
          setMotorSpeed(prop,0);
          stap = true;
          break;
        }
        setMotorSpeed(prop, i);
        sleep(400);
      }
```

Senses if the reflexive value of the light sensor detects an object

If not, for loop to speed up the propeller

If the light sensor detects an object, the propeller would stop rotating

Object Detection - cont.

```
if (stap)
{
  setMotorSpeed(prop,0);
  for (int a = 1; a < 11; a++)
  {
    setMotorSpeed(rot_prop,10);
    sleep(400);
    if (SensorValue[objdet] <10)
    {
      setMotorSpeed(rot_prop,0);
      stap = false;
      break;
    }
    else if (a==10)
    {
      for (int b = 1; b<21; b++)
      {
        setMotorSpeed(rot_prop,-10);
        sleep(200);
        if (SensorValue[objdet] < 10)
        {
          setMotorSpeed(rot_prop, 0);
          stap = false;
          break;
        }
      }
    }
```

If the object was detected, propeller used for rotating the submarine will turn clockwise

Once the object is gone, the loop terminates

If object did not vanish, the robot will turn counter clockwise

Critiques

1

Syringe + glue gun= leak?
→
Superglue on top

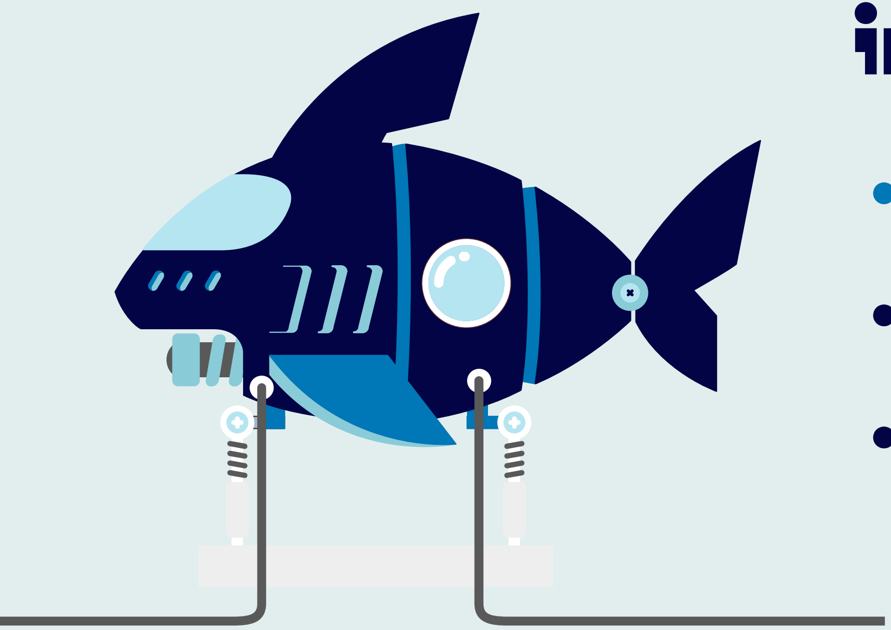
2

plastic cylinder instead of a glass jar for holes & weight
→
Used plastic cylinder

3

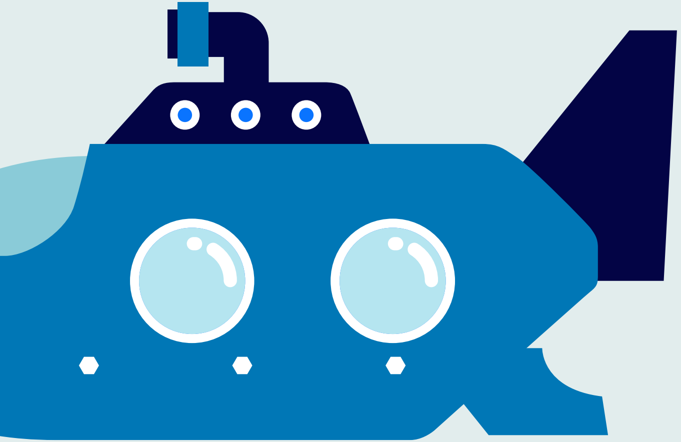
Not air-tight?
→
Added napkins

Areas of improvement



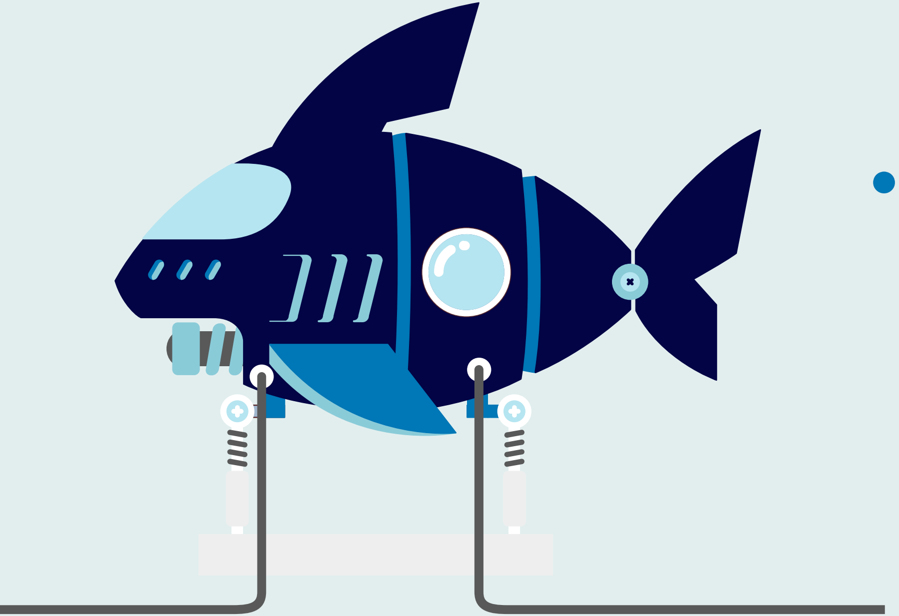
- Putting them all together
- Stronger magnets for propellers
- Better depth control & weights

Next Steps



- Flashlight capabilities
- Recording capabilities
- Remote control
- Improved automation

Practical purpose



- Underwater exploration
    - Environmental intelligence
    - New discoveries
        - Biomimicry

What we learned

- Mechanics of submarine
- Using gears
- Importance of thorough **preparation**
- Persistence & teamwork

THANKS!

Do you have any questions?

Link to Demo Video

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